

Day 1: Why Cloud + A Safe AWS Account

Saturday, 10 October 2026 · 7:00-9:00 PM IST · CLF-C02 tasks 1.1 · 1.2 · 1.4 · 2.3 · 4.2 · Trainer: Gangadhar Ure

Week 1 - Day 1: Why Cloud + A Safe AWS Account

Live class: Saturday 10 Oct 2026, 7:00-9:00 PM IST Exam focus: Domain 1 (Cloud Concepts), Domain 2 (Security), Domain 4 (Billing) Exam tasks: 1.1 · 1.2 · 1.4 · 2.3 · 4.2

Pre-work: create your AWS account before the class (Lab 1 in [03-account-setup-console-tour.md](#)) and keep an authenticator app ready on your phone.

Practice Sequence

Seq	Practice	File
01	Why cloud and the problems it solves	01-cloud-foundations.md
02	Six pillars, certifications and careers	02-well-architected-and-careers.md
03	Create your account and tour the Console	03-account-setup-console-tour.md
04	Root MFA and budget alert	04-account-security-lab.md

PDF guide: [Day 1 Learner Guide](#)

By The End Of Today

- You can explain five problems the cloud solves and the six advantages of cloud computing.
- You can explain high availability, elasticity and agility, and basic cloud economics (fixed vs variable cost, BYOL, rightsizing).
- You know IaaS, PaaS and SaaS, and the six Well-Architected pillars.
- You know the CLF-C02 exam format.
- You can sign in, switch Regions and find your Free Tier or credits page.
- Your root user has MFA and you have a budget alert.
- You can name tasks only the root user can do, and tell Budgets, Cost Explorer and Pricing Calculator apart.

Today's Deliverables

- Screenshot of Console Home with the Region visible (account ID hidden)
- Screenshot showing root MFA assigned (no QR, no codes)
- Screenshot of the budget list
- Account security checklist ticked in `notes.md`
- Career path notes in `notes.md` (exam details, job skills) from [02-well-architected-and-careers.md](#)
- Day 1 LinkedIn post: see [10-linkedin-post.md](#)

Self-Check: Exam-Style Questions

Answer first, then open the answer.

1. A startup wants to avoid buying servers upfront and pay only for what it uses. Which cloud benefit is this?

A. Economies of scale · B. Trade fixed expense for variable expense · C. Go global in minutes · D. High availability

▼ Answer

B. Paying only for usage instead of buying upfront is trading fixed (capital) expense for variable expense.

2. An app adds servers during a sale and removes them afterwards, automatically. What is this called?

A. Agility · B. Elasticity · C. Durability · D. Rightsizing

▼ Answer

B. Elasticity: resources grow and shrink with demand.

3. Which Well-Architected pillar focuses on recovering quickly from failures?

A. Operational Excellence · B. Performance Efficiency · C. Reliability · D. Security

▼ Answer

C. Reliability.

4. Which TWO actions best protect the root user? (Choose two.)

A. Turn on MFA · B. Create root access keys for the CLI · C. Use an IAM user or role for daily work · D. Share the root password with your team

▼ Answer

A and C. Never create root access keys and never share root credentials.

5. Which task can ONLY the root user perform?

A. Launch an EC2 instance · B. Create an S3 bucket · C. Close the AWS account · D. Create an IAM group

▼ Answer

C. Close the AWS account.

6. You want an email when monthly spend crosses USD 5. Which service?

A. AWS Cost Explorer · B. AWS Budgets · C. AWS Pricing Calculator · D. AWS Artifact

▼ **Answer**

B. AWS Budgets. It alerts you; it does not stop spending.

Day 1 - Cloud Foundations

Goal: understand what the cloud is and which real problems it solves.

Exam tasks: 1.1 (benefits of the AWS Cloud), 1.4 (cloud economics), 3.1 (deployment models)

Before You Start

Write answers in your own words before reading:

1. What is cloud computing? -->
2. Why would a company use cloud instead of its own servers? -->
3. Name one service you already use that runs on the cloud. -->

The Story

Your college fest website gets 50 visitors a day for 360 days, then 50,000 on registration day. With your own servers you buy for the peak, wait weeks, and pay all year. With cloud you rent what you need in minutes and pay only for what you use.

Five Problems The Cloud Solves

Problem	Without cloud	With AWS	Service (later)
Scaling (autoscaling)	Buy for the peak	Capacity grows and shrinks with traffic	EC2 Auto Scaling, Elastic Load Balancing
Cost	Large upfront spend	Pay for what you use	Budgets, Cost Explorer
Availability	One data centre fails, site down	Spread across Availability Zones	Multi-AZ, Route 53, CloudFront
Security	You secure everything alone	Shared responsibility	IAM, KMS, Security Groups
Reliability	Manual backups	Automated backups and replication	S3, RDS, AWS Backup

Six Advantages Of Cloud Computing

1. Trade fixed expense for variable expense.
2. Benefit from massive economies of scale.
3. Stop guessing capacity.
4. Increase speed and agility.
5. Stop spending money running and maintaining data centres.
6. Go global in minutes.

Four Words The Exam Loves

Word	Meaning	Fest example
High availability	The app keeps running when one part fails	Site stays up if one data centre has a problem
Elasticity	Resources grow and shrink automatically with demand	2 servers normally, 20 on registration day, back to 2
Scalability	The system can handle more load by adding resources	Ready to grow when the fest gets bigger every year
Agility	Try new things fast and cheaply	Launch a test feature in minutes, delete it if it fails

Global reach is a benefit too: deploy close to users anywhere in minutes.

Cloud Economics

Idea	On-premises	AWS Cloud
Fixed vs variable cost	Big upfront spend (CapEx) on servers, buildings, power, cooling, staff	Pay for what you use (OpEx)
Hidden on-prem costs	Hardware refresh, electricity, cooling, floor space, security guards, data centre staff	Included in the service price
Economies of scale	Your small volume, your price	AWS buys at huge volume and passes savings on
Rightsizing	Hard: you already bought the server	Pick the right size, change it any time
Automation	Manual setup, slow, error-prone	Scripts and IaC: faster, repeatable, fewer mistakes
Licensing	You own licences	BYOL (bring your own licence) or licence included in the price

Exam pointer:

- "Reduce upfront cost" or "trade capital expense for variable expense" = cloud economics.
- "Resources match demand automatically" = elasticity.
- "Already own Windows or Oracle licences" = BYOL.
- "Instance is too big for the workload" = rightsizing.

Service Models

- **IaaS**: you manage OS and apps (Amazon EC2).
- **PaaS**: you manage app and data; the platform is managed (Elastic Beanstalk, RDS).
- **SaaS**: you just use it (Gmail, Microsoft 365).

Deployment Models

- Cloud, on-premises (private cloud), and hybrid (both together).

Exam pointer:

- Pay-as-you-go and no upfront cost point to cloud benefits.
- Own data centre plus AWS together means hybrid.
- You manage the OS means IaaS.

Practice - Explore The Official AWS Pages

Practice in this order:

1. Open [What is cloud computing?](#). Find the section on benefits.
2. Open [Types of cloud computing](#). Find IaaS, PaaS and SaaS.
3. Open [AWS customer stories](#). Filter by one industry you like (for example Education, Media or Financial Services).
4. Open one story. Read the headline and the results.

Add to your `notes.md` :

```
Cloud in one line (my words):  
One benefit I found on the AWS page:  
Customer story I read (company + one result it mentions):
```

Expected result: three short lines in your notes, written in your own words.

Official Links

Topic	Link
What is cloud computing	https://aws.amazon.com/what-is-cloud-computing/
Types of cloud computing (IaaS, PaaS, SaaS)	https://aws.amazon.com/types-of-cloud-computing/
Six advantages of cloud computing	https://docs.aws.amazon.com/whitepapers/latest/aws-overview/six-advantages-of-cloud-computing.html
AWS customer stories	https://aws.amazon.com/solutions/case-studies/
AWS Pricing overview	https://aws.amazon.com/pricing/

Day 1 - Six Pillars, Certifications and Careers

Exam task: 1.2 (design principles of the AWS Cloud)

The Six Pillars Of The AWS Well-Architected Framework

Pillar	Question it asks	Benefit	Example
Operational Excellence	Can we run, monitor and improve it?	Fewer outages	Automate deployments, monitor
Security	Is data and access protected?	Trust, compliance	Least-privilege IAM, encryption
Reliability	Does it recover and keep working?	Less downtime	Multi-AZ, backups
Performance Efficiency	Do we use the right resources?	Fast, no waste	Right instance type, caching
Cost Optimization	Do we pay only for value?	Lower bills	Right-size, turn off idle
Sustainability	Do we minimise environmental impact?	Lower energy and cost	Efficient resources, scale down

Exam pointer: "recover from failure" = Reliability. "Reduce spend" = Cost Optimization. "Protect data" = Security.

Common confusion:

- Performance Efficiency is about using the **right** resources well (speed). Cost Optimization is about **not paying** for waste.
- Operational Excellence is about **running and improving** the workload. Reliability is about **recovering** from failure.
- Sustainability is about **energy and environmental impact**, often by removing idle resources.

Certification Path

- Foundational: **Cloud Practitioner (CLF-C02)** (our goal), AI Practitioner.
- Associate: Solutions Architect, Developer, SysOps/CloudOps Engineer, Data Engineer.
- Professional and Specialty: senior roles and specialisation. Check aws.amazon.com/certification for the current list, prices and renewal rules.

The CLF-C02 Exam At A Glance

Item	Detail
Questions	65 (50 scored + 15 unscored; you cannot tell which)
Time	90 minutes
Question types	Multiple choice (one answer) and multiple response (two or more answers)
Pass mark	700 out of 1000 (scaled score); you only need to pass overall

Item	Detail
Guessing	No penalty, so never leave a question blank
Domains	Cloud Concepts 24% · Security and Compliance 30% · Cloud Technology and Services 34% · Billing, Pricing and Support 12%

Roles That Use AWS Skills

Cloud Support Associate, Cloud Engineer, DevOps Engineer, Solutions Architect, Data Engineer, Security Engineer.

Employers look for hands-on skills plus a certification. That is why every week has a practical and a GitHub submission: it becomes your portfolio.

Lab - Explore Your Career Path

Practice in this order:

1. Open [AWS Certification](#). Find the four levels: Foundational, Associate, Professional, Specialty.
2. Open [AWS Certified Cloud Practitioner](#). Note the exam code, time and number of questions.
3. On the same page, open the **exam guide**. Find the four domains and their weights.
4. Open [AWS Skill Builder](#). Sign up for free. Search `Cloud Practitioner` and find one free course.
5. Open LinkedIn Jobs or Naukri. Search `AWS fresher`, location `India`.
6. Open two job posts. Write down the skills they ask for.

Add to your `notes.md` :

```
Exam code, time, number of questions:
Four exam domains:
Free Skill Builder course I found:
Job 1 skills:
Job 2 skills:
One skill I will build in this series:
```

Expected result: your own small career plan in `notes.md`. Do not post salary or personal details.

If you get stuck, write what you completed:

```
I found the exam details. I could not sign up for Skill Builder yet.
```

Reflection (add to your notes)

Which role interests you most and why? Which pillar surprised you?

Official Links

Topic	Link
Six pillars of the Well-Architected Framework	https://docs.aws.amazon.com/wellarchitected/latest/framework/the-pillars-of-the-framework.html
AWS Well-Architected overview	https://aws.amazon.com/architecture/well-architected/
AWS Certification (all levels)	https://aws.amazon.com/certification/
AWS Certified Cloud Practitioner (exam page and guide)	https://aws.amazon.com/certification/certified-cloud-practitioner/
AWS Skill Builder (free training)	https://skillbuilder.aws/
AWS Training and Certification	https://aws.amazon.com/training/

Day 1 - Account Setup and Console Tour

Exam tasks: 3.1 (ways to access AWS), 4.3 (AWS Support plans)

Practice Rule

Use your **own** AWS account. Never share your email, password, OTP, card or account ID.

Practice in this order:

1. Create your AWS account (before the live class).
2. Sign in as the root user.
3. Tour the Console.
4. Take one safe screenshot.

Lab 1 - Create Your AWS Account

Please do this **before the Day 1 live class**. Verification and activation can take time. Skip if you already have an account.

1. Open aws.amazon.com and choose **Create an AWS Account**.
2. Enter your personal email and an account name, for example `cloudadhar-learning`.
3. Open your email, copy the verification code and paste it.
4. Create a strong, unique root password. Save it in a password manager.
5. Choose the **Free plan** for learning. It gives credits for a limited time. Check the Billing page for current terms.
6. Choose **Personal**. Enter your contact details. Accept the agreement.
7. Add a payment method. AWS uses it to verify identity. Never share these details.
8. Verify your phone by SMS or voice call.
9. Choose the **Basic support** plan. It is free.
10. Wait for activation. It takes minutes, sometimes up to 24 hours.
11. Go to the [AWS Console](#). Choose **Root user** and sign in.

Expected result: you see **Console Home**.

Screens and plan names can change. Follow the on-screen text.

Lab 2 - Console Tour

Test:

1. Look at the top right. Find the **Region** name.
2. Click it and choose **Asia Pacific (Mumbai) ap-south-1**.
3. Click the search bar at the top. Type `EC2` and open it. Click the AWS logo to go back.
4. Search `S3` and open it. Go back.
5. Switch the Region to **US East (N. Virginia) us-east-1**. Notice the Region name change. Switch back to **Mumbai**.
6. Search `Billing` and open **Billing and Cost Management**.
7. Find **Free Tier** or **Credits** in the left menu. Look at what is used and what is left.

8. Click **Services** (or the grid icon). Find one service in each group: Compute, Storage, Database, Networking, Security.

Expected result: you can say which Region you are in and where to find your credits.

Add to your `notes.md` :

```
My Region:
One service each – Compute: / Storage: / Database: / Networking: / Security:
What surprised me in the Console:
```

Deliverables

- Screenshot of Console Home with the Region visible (account ID hidden)
- One sentence: what surprised you in the Console?

If You Get Stuck

Problem	Try this
No OTP or SMS	Wait 2 minutes, retry, or choose voice call
Card verification fails	Try another card or bank. Never post card details in chat
Account "pending activation"	Normal. Wait up to 24 hours, then sign in again
Console in another language	Gear icon (Settings) → Language

Submit what you completed with a note:

```
My account is still pending activation. I watched the Console tour and will finish tomorrow.
```

Next: secure your account right away in [04-account-security-lab.md](#).

Official Links

Topic	Link
Create and activate an AWS account	https://repost.aws/knowledge-center/create-and-activate-aws-account
Getting started with your AWS account	https://docs.aws.amazon.com/accounts/latest/reference/getting-started.html
AWS Free Tier	https://aws.amazon.com/free/
AWS Free Tier plans and credits	https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/free-tier-plans.html
AWS Management Console guide	https://docs.aws.amazon.com/awsconsolehelpdocs/latest/gsg/what-is.html
AWS Support plans	https://aws.amazon.com/premiumsupport/plans/

Day 1 - Account Security Lab

Goal: create a safe AWS account foundation for the next five weeks.

Exam tasks: 2.3 (root user protection, MFA), 4.2 (AWS Budgets)

Why The Root User Is Special

The root user is the email you signed up with. It has full access to everything and cannot be limited by IAM policies.

Some tasks only the root user can do, for example:

- Change account settings: account name, root email, root password
- Close the AWS account
- Change or cancel the AWS Support plan
- Restore permissions when the only IAM admin is locked out
- Turn on IAM user access to the Billing console

Everything else should be done with an IAM user or role (Week 2).

Exam pointer: "protect the root user" = turn on MFA, do not create root access keys, do not use root for daily work, use a strong unique password.

Practice Rule

Only the root user does this lab. Do not create any servers or storage today.

Practice in this order:

1. Install an authenticator app on your phone.
2. Turn on MFA for the root user.
3. Test MFA by signing in again.
4. Check that there are no root access keys.
5. Create a budget alert.

Before You Start

You need:

- Your root email and password
- A phone with an authenticator app (Google Authenticator, Microsoft Authenticator or similar)
- Access to your email

Lab 1 - Secure The Root User

Create:

- MFA device name: `root-mfa-phone`

Steps:

1. Sign in to the [AWS Console](#) as the **Root user**.
2. Click your account name at the top right. Choose **Security credentials**.
3. Find the **Multi-factor authentication (MFA)** section. Click **Assign MFA device**.
4. Device name: `root-mfa-phone`.
5. Choose **Authenticator app**. Click **Next**.
6. Click **Show QR code**. Open your authenticator app and scan it. Do **not** screenshot the QR.
7. Type the 6-digit code from the app into **MFA code 1**.
8. Wait for the code to change (about 30 seconds). Type the new code into **MFA code 2**.
9. Click **Add MFA**.

Test:

1. Click your account name → **Sign out**.
2. Sign in again as the root user.
3. AWS asks for an MFA code. Type the code from your app.

Expected result: you are signed in, and the MFA device `root-mfa-phone` is listed under **Security credentials**.

Check access keys:

1. On the same **Security credentials** page, find **Access keys**.
2. It should say there are no access keys. Do not create any.
3. If you find an old key you do not use, delete it. If you are unsure, ask in the community first.

Stop using root for daily work. We create a safer IAM user in Week 2.

Deliverables:

- Screenshot showing root MFA assigned (no QR, no codes, account ID hidden)
- Short note: why root should not be used daily

Lab 2 - Budget Alert

Create:

- Budget name: `cloud-unlock-learning-budget`
- Amount: USD 5 (or the Zero spend template)

Steps:

1. Search `Budgets` in the Console. Open **Billing and Cost Management** → **Budgets**.
2. Click **Create budget**.
3. Choose **Use a template (simplified)**.
4. Choose **Zero spend budget**, or **Monthly cost budget** with amount `5`.
5. Budget name: `cloud-unlock-learning-budget`.
6. Email recipients: your email.
7. Click **Create budget**.

Test:

1. Go back to **Budgets**.
2. Find `cloud-unlock-learning-budget` in the list.

Expected result: the budget appears in the list with your amount.

Remember: budgets **alert** you. They do **not** stop spending. Alert emails can arrive late, so always clean up after labs.

Deliverables:

- Screenshot of the budget list
- Short note: why monitor cost from day one (budgets alert; they do not stop spending)

Budgets Or Cost Explorer?

Tool	Use it to	Think of it as
AWS Budgets	Get an alert before or when spend crosses a limit	Smoke alarm
AWS Cost Explorer	See and analyse past spend and forecasts	Bank statement
AWS Pricing Calculator	Estimate cost before you build	Shopping quote

Account Security Checklist

Tick these in your `notes.md` (do not share any IDs):

- ☐ Root user has MFA
- ☐ No root access keys exist
- ☐ Root password is strong and stored safely (for example in a password manager)
- ☐ Budget alert created with my email
- ☐ I know which Region I work in (Mumbai `ap-south-1`)
- ☐ I know where to find Billing and Free Tier or credits

If You Get Stuck

Problem	Try this
"Codes not valid"	Set your phone time to automatic. Wait for a new code before typing the second one
Lost the phone with MFA later	Use Troubleshoot MFA on the sign-in page (email and phone check)
Budgets page shows an error	Make sure you signed in as the root user

Add a screenshot of where you got stuck, write 2-3 lines, and ask in the community. Do not skip the whole week.

I turned on MFA. I got stuck creating the budget because the page showed an error.

Safety Note

Do not share root email, account ID, access keys, MFA QR code, OTP, payment details or detailed billing.

Official Links

Topic	Link
Root user best practices	https://docs.aws.amazon.com/IAM/latest/UserGuide/root-user-best-practices.html
Tasks that need the root user	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_root-user.html
Turn on MFA for the root user	https://docs.aws.amazon.com/IAM/latest/UserGuide/enable-virt-mfa-for-root.html
MFA in AWS	https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa.html
AWS Budgets	https://docs.aws.amazon.com/cost-management/latest/userguide/budgets-managing-costs.html
AWS Cost Explorer	https://docs.aws.amazon.com/cost-management/latest/userguide/ce-what-is.html
AWS Pricing Calculator	https://calculator.aws/

github.com/cloudadhar/cloud-unlock-aws-challenge · youtube.com/@CloudAdhar · Use synthetic data only. Never share account IDs, access keys, MFA QR codes, OTPs or card details.